POLITECNICO MILANO 1863	052820 SOUND ANALYSIS, SYNTHESIS AND PROCESSING Module 1: DIGITAL AUDIO ANALYSIS AND PROCESSING Academic Year 2025/2026	DAAP Homework Linear Predictive Coding

Assignment: The talk box effect

The talk box is a classic audio effect widely used in music production, particularly in genres such as rock and funk. Although the first prototypes date back to the late 30s, it became popular starting from the 70s and throughout the 80s. You can listen to the effect in this video and in several well-known songs. The effect works by routing a sound source (e.g., a guitar or synthesizer) through a tube into the performer's mouth. By shaping the vocal tract, the performer modulates the spectral characteristics of the sound, producing a speech-like effect.

From a signal processing perspective, this device can be interpreted as a **source–filter model**, where the input signal acts as an **excitation signal** and the vocal tract acts as a **time-varying filter**.

In this assignment, you will explore how to recreate a **digital version of this effect**. Rather than physically modeling the acoustic system, you will implement a speech-driven spectral shaping system using the adaptive filtering techniques introduced in the course.

In particular, your system should rely on **linear prediction** to model the spectral characteristics of speech and transfer them to an excitation signal (e.g., a guitar). The final effect should resemble a "talking" instrument driven by the speech signal.

The overall objective is to design and implement a system that

- extracts a time-varying spectral model from a speech signal;
- applies the speech model to the residual of the excitation signal;
- produces an output signal in which the excitation signal is shaped by the speech characteristics.

The system must operate frame-by-frame and reconstruct a continuous output signal.

Minimum requirements

Your implementation has to provide the following features:

- Frame-based processing (with windowing and overlap-and-add reconstruction);
- LPC-based analysis;
- LPC-based resynthesis with excitation signal;
- Generate at least 3 audio examples (one example is provided).

You are free to make your own design choices, including:

- LPC order for both the speech and music signals;
- Frame length and hop size;
- Adoption of different types of excitation signals;
- Any additional processing steps.

All design choices must be clearly explained in the report and justified based on your own experimentation. There is no single correct solution; different design choices may lead to different perceptual results.

Analysis of the system

You are expected to evaluate how the design of the parameters affects the output and discuss the limitations of your approach. In particular, you are required to vary **at least two parameters** of your model (e.g., LPC order, and excitation type) and discuss how different parameter choices influence the perceptual quality and intelligibility of the output signal. Support your discussion with specific examples.

You are asked to submit the source codes and all accessory files required to reproduce your results. In addition, provide a report in PDF format that must include the system description with implementation details, it must explain the design choices and the experimental analysis. Please include figures in order to support your explanation.

Use of AI/LLM Tools

The use of AI-based tools (e.g., ChatGPT, Claude etc.) is allowed (although not recommended), and **must be declared** in the report.
If employed, their usage must be documented in the report, including:
- how the tool was used
- examples of the relevant prompts
- how results were verified and modified

You must be able to explain all parts of your implementation.
Undocumented or uncritical use of AI tools will negatively impact the evaluation.

Instructions

- Groups of at **most 2 people** are allowed.
- The project must be implemented in MATLAB.
- Each group should upload the code on WeBeep as a **single zip file**. One student will submit a zip file for the entire group; **do not upload the same HW twice**; please use `.zip` format
- The zip file should be named with the surnames of all group members, e.g., Mario Rossi and Maria Bianchi will upload a file named `DAAP_HW1_Rossi_Bianchi.zip`
- Upload the required file using the **WeBeep platform** in the delivery folder.
- The instructor reserves the right to ask for a short oral discussion of the submitted work. During this discussion, students may be asked to explain and justify their implementation and design choices.